

Lecture 3:

Descriptive Statistics

Introduction

Broadly classified into two categories, namely

- Descriptive Statistics
- Inferential Statistics

Descriptive Statistics

Descriptive statistics are used to summarize and describe the main features of a dataset.

- They aim to provide a clear and concise summary of the data, typically through measures such as central tendency (mean, median, mode), dispersion (range, variance, standard deviation), and visual representations like histograms, bar charts, and scatterplots.
- Descriptive statistics focus on organizing, displaying, and summarizing data in a meaningful way without making any inferences or generalizations beyond the data itself.

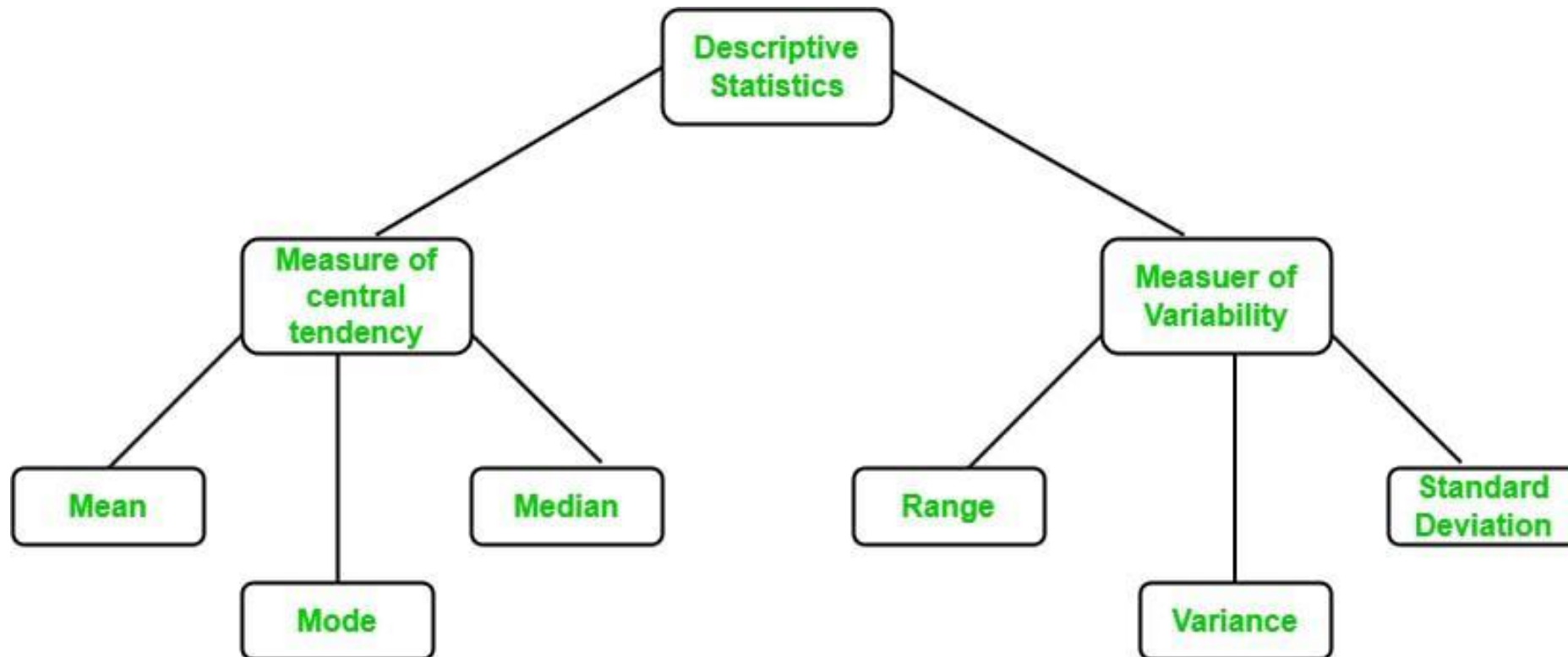
Inferential Statistics

Inferential statistics involve making inferences or conclusions about a population based on data collected from a sample of that population.

- The primary goal of inferential statistics is to generalize findings from a sample to a larger population and to test hypotheses or make predictions.
- Techniques used in inferential statistics include hypothesis testing, estimation, regression analysis, and various tests of significance such as t-tests, ANOVA, chi-square tests, etc.

Types of Descriptive Statistics

- Measure of Central Tendency
- Measure of Variability



Mean

The formulas for the sample mean and the population mean only differ in mathematical notation. Population attributes use capital letters while sample attributes use lowercase letters.

Population mean

Population mean formula	Explanation
$\bar{X} = \frac{\sum X}{N}$	• $\bar{X}$ = population mean • $\sum X$ = sum of each value in the population • N = number of values in the population

The population mean can also be denoted as μ .

Sample mean

Sample mean formula	Explanation
$\bar{x} = \frac{\sum x}{n}$	• $\bar{x}$ = sample mean • $\sum x$ = sum of each value in the sample • n = number of values in the population

The sample mean is also referred to as M .

Outlier effect on the mean

- **Outliers** are extreme values that differ from most values in the data set. Because all values are used in the calculation of the mean, an outlier can have a dramatic effect on the mean by pulling the mean away from the majority of the values.
- Let's see what happens to the mean when we add an outlier to our data set.

Effect of outlier

Data set

Cost of dinner for two (USD)

42 13 31 87 24 58 76 69 230

Step 1: Find the sum of the values by adding them all up

Formula	Calculation
$\sum x$	$42 + 13 + 31 + 87 + 24 + 58 + 76 + 69 + 230 = 630$

Step 2: Divide the sum by the number of values

Formula	Calculation
$\bar{x} = \sum x \div n$	$n = 9$ $\sum x = 630$ $\bar{x} = 630 \div 9 = 70$

Median

- The median is the value in the middle of a data set, meaning that 50% of data points have a value smaller or equal to the median and 50% of data points have a value higher or equal to the median.

Imagine that a top running athlete in a typical 200-metre training session runs in the following times: 26.1 seconds, 25.6 seconds, 25.7 seconds, 25.2 seconds, 25.0 seconds, 27.8 seconds and 24.1 seconds. How would you calculate his median time?

Let's start with arranging the values in increasing order:

Table 4.4.2.1
Rank associated with
each value of 200-
meter running times

Rank	Times (in seconds)
1	24.1
2	25.0
3	25.2
4	25.6
5	25.7
6	26.1
7	27.8

There are $n = 7$ data points, which is an uneven number. The median will be the value of the data points of rank

$$(n + 1) \div 2 = (7 + 1) \div 2 = 4.$$

The median time is 25.6 seconds.

Now suppose that the athlete runs his eighth 200-metre run with a time of 24.7 seconds. What is his median time now?

Table 4.4.2.2

**Rank associated with
each value of 200-
meter running times,
updated**

Rank	Times (in seconds)
1	24.1
2	24.7
3	25.0
4	25.2
5	25.6
6	25.7
7	26.1
8	27.8

There are now $n = 8$ data points, an even number. The median is the mean between the data point of rank

$$n \div 2 = 8 \div 2 = 4$$

and the data point of rank

$$(n \div 2) + 1 = (8 \div 2) + 1 = 5$$

Therefore, the median time is $(25.2 + 25.6) \div 2 = 25.4$ seconds.

Mode

The mode is the value that appears the most often in a data set and it can be used as a measure of central tendency

During a hockey tournament, Audrey scored 7, 5, 0, 7, 8, 5, 5, 4, 1 and 5 points in 10 games. After summarizing the data in a frequency table, you can easily see that the mode is 5 because this value appears the most often in the data set (4 times). The mode can be considered a measure of central tendency for this data set because it's unique.

Table 4.4.3.1
Number of games by the number of points scored

Number of points scored	Frequency (number of games)
0	1
1	1
4	1
5	4
7	2
8	1
0 true zero or a value rounded to zero	

Remarks on Mode

- Do mode always exist a sequence ?
- When we can use the mode ?

Remarks on Mean

- When we cannot use the mean
- What are the advantages of the mean over median

Remarks on the Median

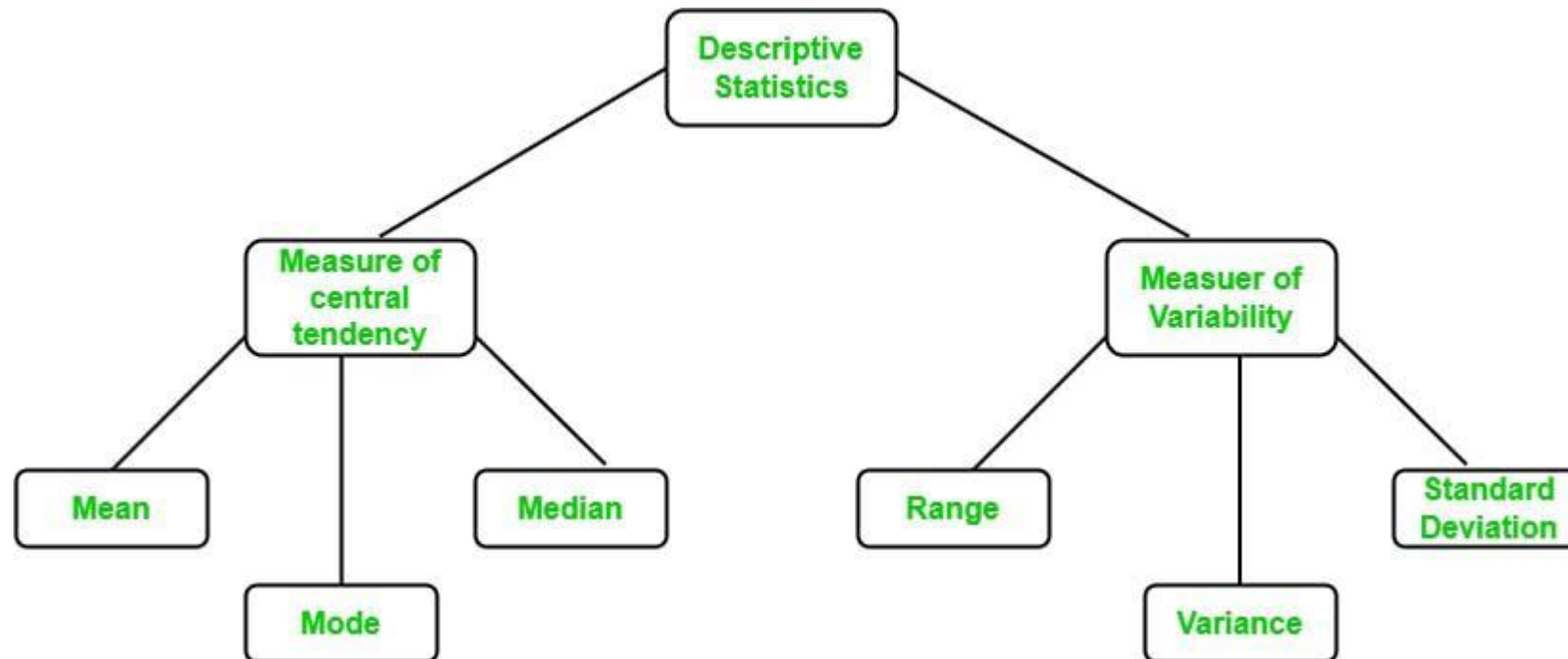
- When to use the median ?

Remarks on Variance

- High Variance is good or bad ?
- Examples

Lecture 2

Review on the last class



Variance

- Variance is a measure of how data points differ from the mean.
- According to Layman, a variance is a measure of how far a set of data (numbers) are spread out from their mean (average) value.

Formula

As we know already, the variance is the square of standard deviation

$$\text{Variance} = (\text{Standard deviation})^2 = \sigma^2$$

The corresponding formulas are hence,

$$\text{Population standard deviation } \sigma = \sqrt{\frac{\sum(X-\mu)^2}{N}}$$

$$\text{Sample standard deviation } s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}}$$

Where X (or x) = Value of Observations

μ = Population mean of all Values

n = Number of observations in the sample set

$\bar{x}$ = Sample mean

N = Total number of values in the population

Standard Deviation

Standard Deviation Formula

The population standard deviation formula is given as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (X_i - \mu)^2}$$

Here,

σ = Population standard deviation

N = Number of observations in population

X_i = i th observation in the population

μ = Population mean

Similarly, the sample standard deviation formula is:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

Here,

s = Sample standard deviation

n = Number of observations in sample

x_i = i th observation in the sample

$\bar{x}$

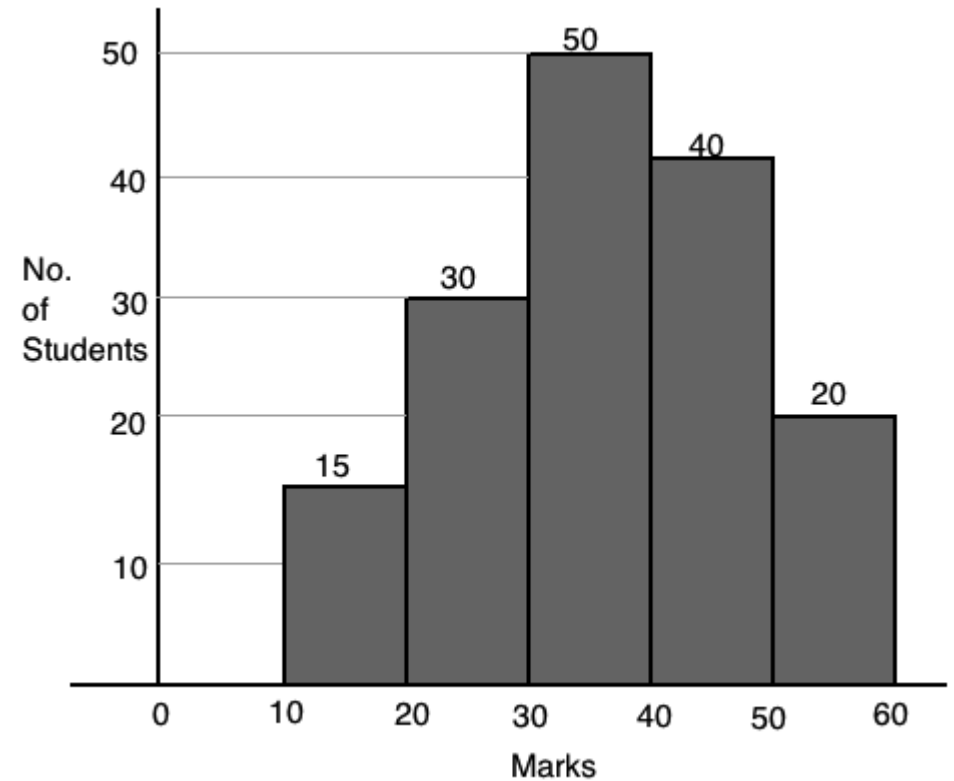
= Sample mean

Distribution of the Data

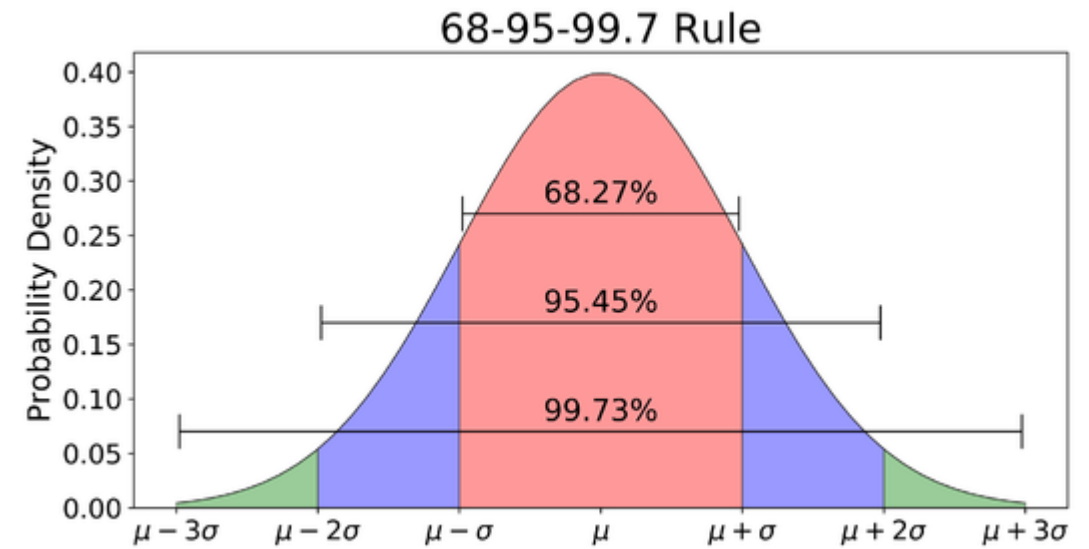
Marks scored:

Draw a histogram for the following data:

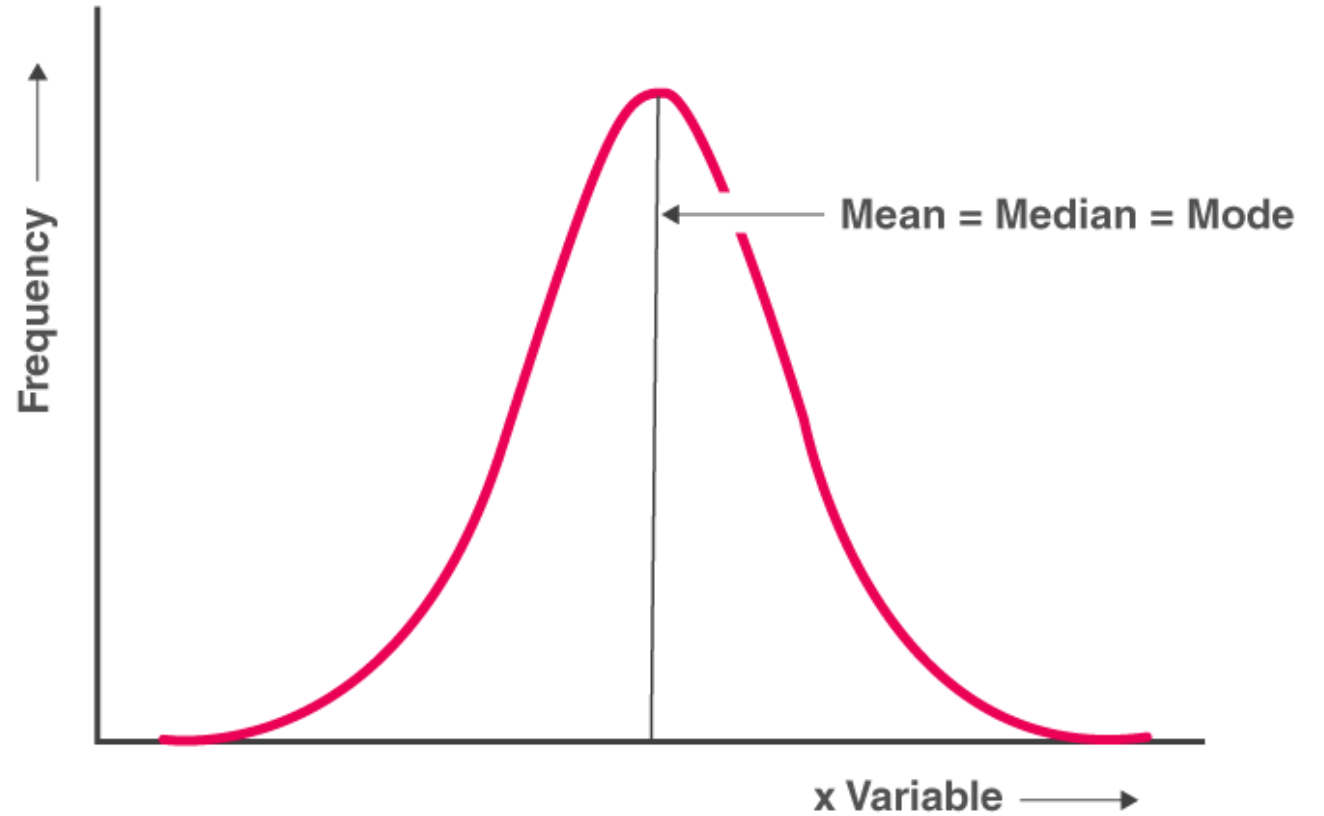
Marks	No. of Students
10 – 20	15
20 – 30	30
30 – 40	50
40 – 50	40
50 – 60	20



Normal Distribution Curve



Frequency Distribution with Symmetrical Frequency Curve



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Mean = Median = Mode

Skewness

- Skewness is a **measure of the asymmetry of the probability distribution** of a real-valued random variable about its mean.
- The skewness value can be positive, zero, negative, or undefined.

Alternate Formula

$$\gamma_1 := \tilde{\mu}_3 = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right]$$

The skewness formula is given by:

$$g = \frac{\sum_{i=1}^n (x_i - \bar{x})^3}{(n-1)s^3}$$

Where,

$\bar{x}$

is the sample mean

x_i

is the i th sample

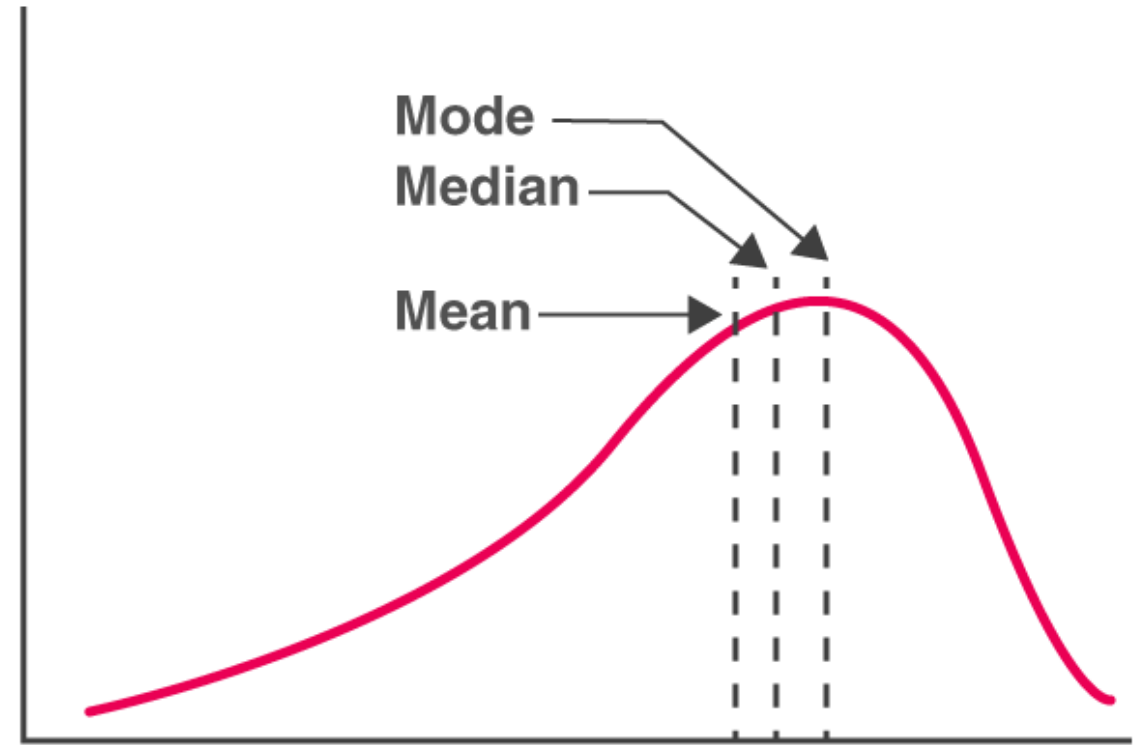
n is the total number of observations

s is the standard deviation

g = sample skewness

For Negatively Skewed Frequency Distribution

- The left tail is longer; the mass of the distribution is concentrated on the right of the figure.
- The distribution is said to be *left-skewed*, *left-tailed*, or *skewed to the left*, despite the fact that the curve itself appears to be skewed or leaning to the right; *left* instead refers to the left tail being drawn out and, often, the mean being skewed to the left of a typical center of the data.
- A left-skewed distribution usually appears as a *right-leaning* curve



Negatively skewed frequency distribution

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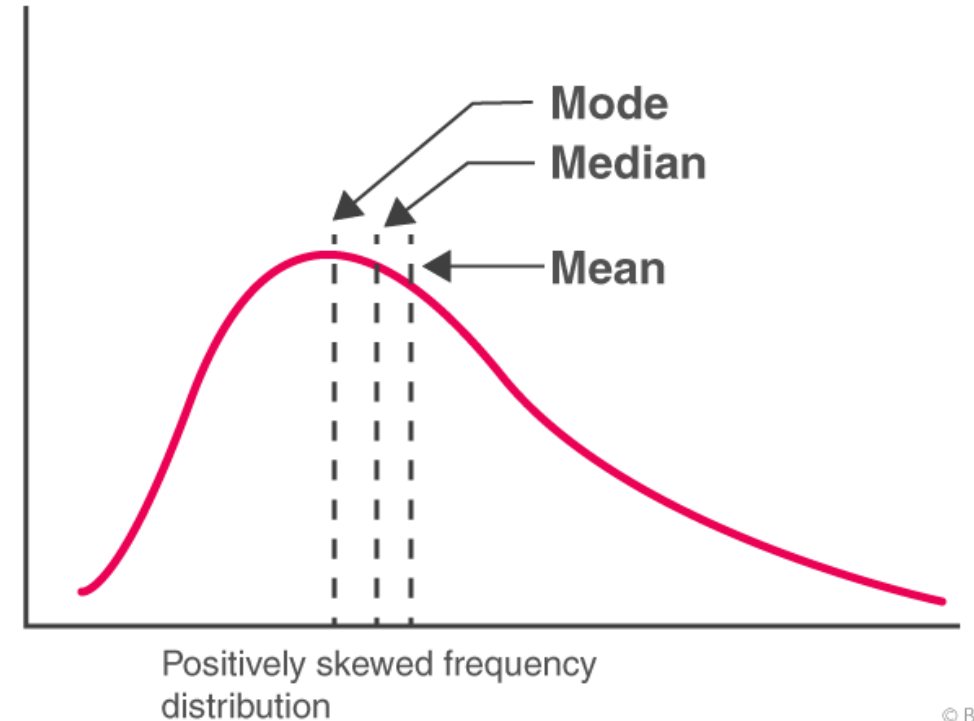
$$\text{Mean} < \text{Median} < \text{Mode}$$

For Positively Skewed Frequency Distribution

The right tail is longer; the mass of the distribution is concentrated on the left of the figure.

The distribution is said to be *right-skewed*, *right-tailed*, or *skewed to the right*, despite the fact that the curve itself appears to be skewed or leaning to the left; *right* instead refers to the right tail being drawn out and, often, the mean being skewed to the right of a typical center of the data.

A right-skewed distribution usually appears as a *left-leaning* curve



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Mean > Median > Mode

Empirical Relationship between Mean, Median and Mode

- In case of a moderately skewed distribution, the difference between mean and mode is almost equal to three times the difference between the mean and median. Thus, the empirical mean median mode relation is given as:

$$\text{Mean} - \text{Mode} = 3 (\text{Mean} - \text{Median})$$

Kurtosis

What is Kurtosis?

Kurtosis is a statistical measure that defines how heavily the tails of a distribution differ from the tails of a normal distribution. In other words, kurtosis identifies whether the tails of a given distribution contain extreme values.

$$\text{Kurt}[X] = \text{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right] :$$

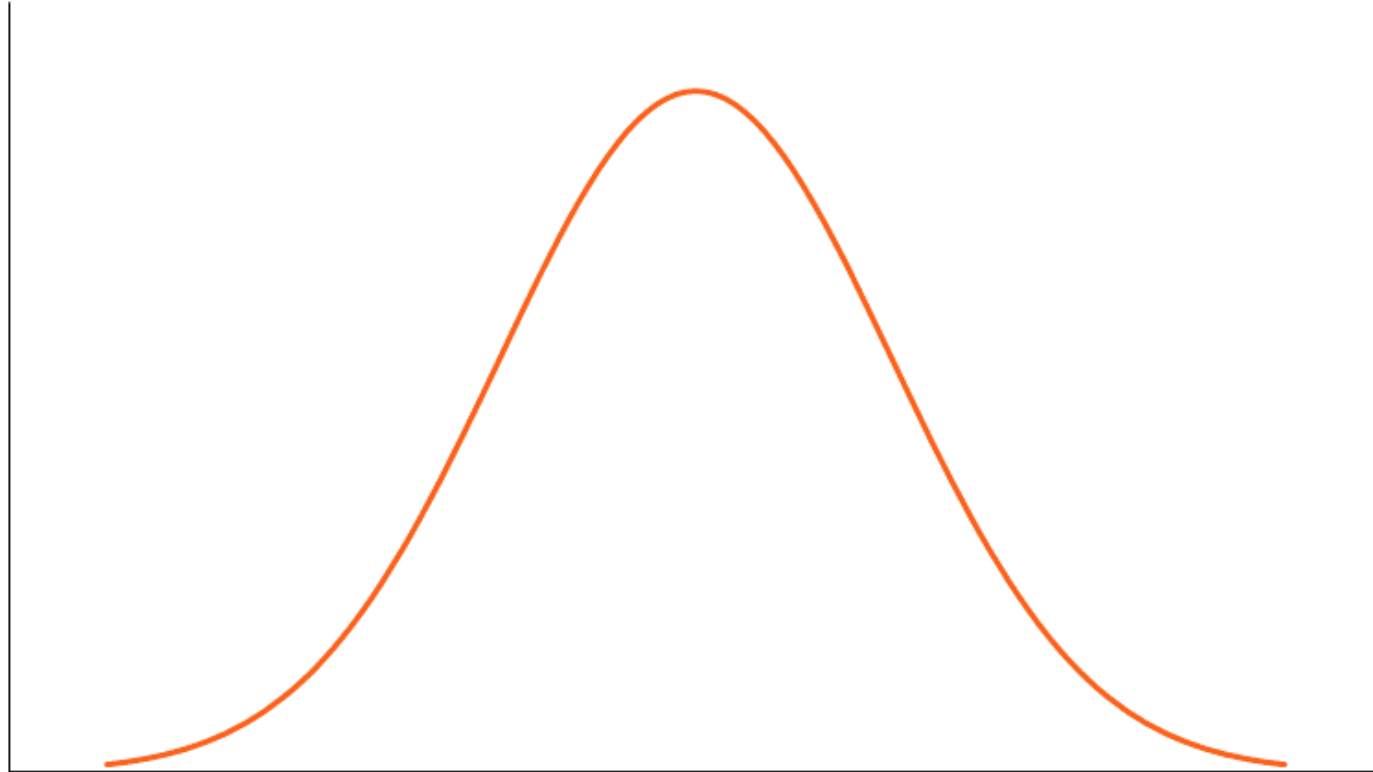
Ref: <https://corporatefinanceinstitute.com/resources/knowledge/other/kurtosis/>

- **What is Excess Kurtosis?**

- Excess kurtosis is a metric that compares the kurtosis of a distribution against the kurtosis of a normal distribution. The kurtosis of a normal distribution equals 3. Therefore, the excess kurtosis is found using the formula below:
- **Excess Kurtosis = Kurtosis - 3**

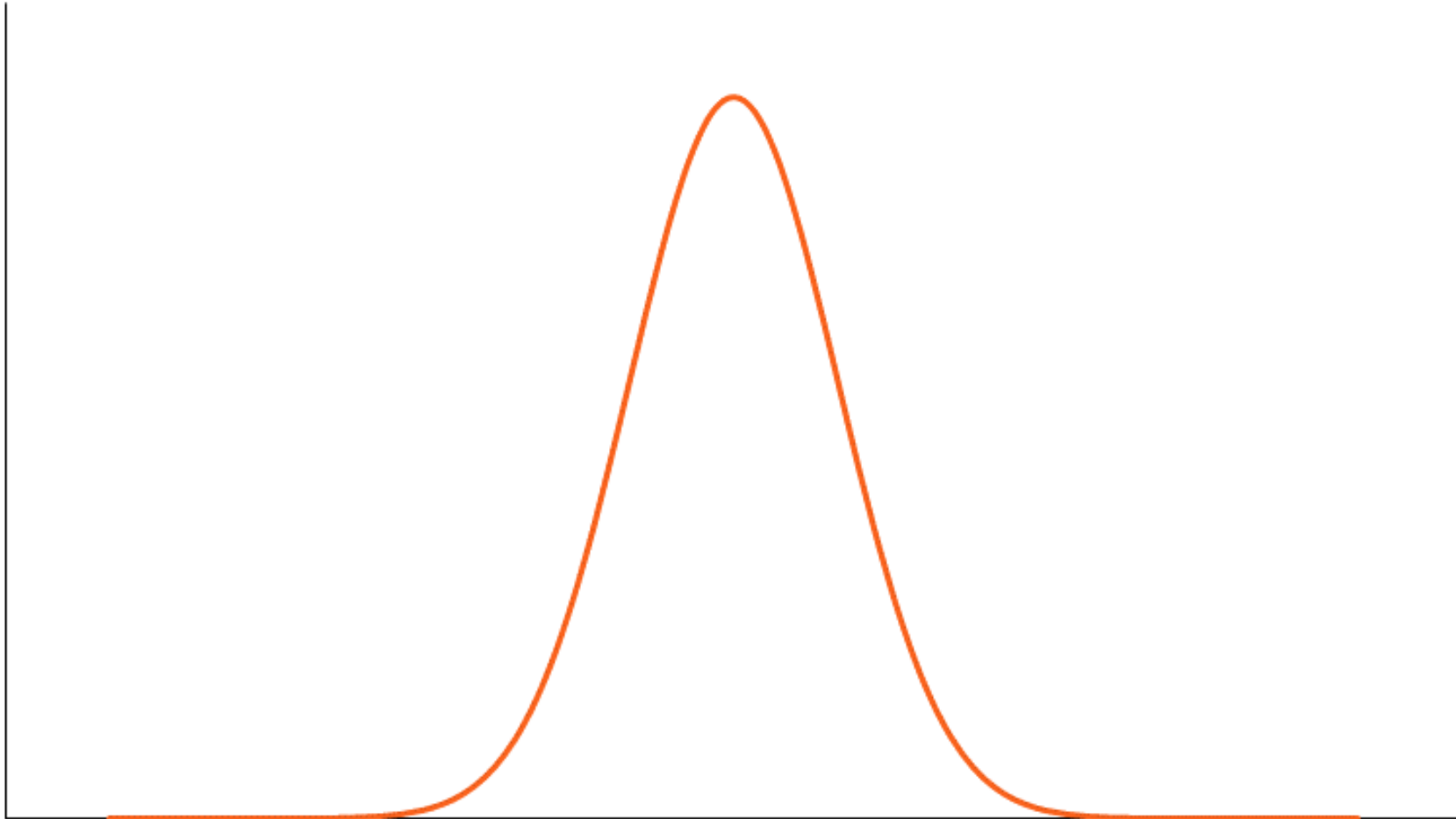
Mesokurtic

- Data that follows a mesokurtic distribution shows an excess kurtosis of zero or close to zero. This means that if the data follows a normal distribution, it follows a mesokurtic distribution.



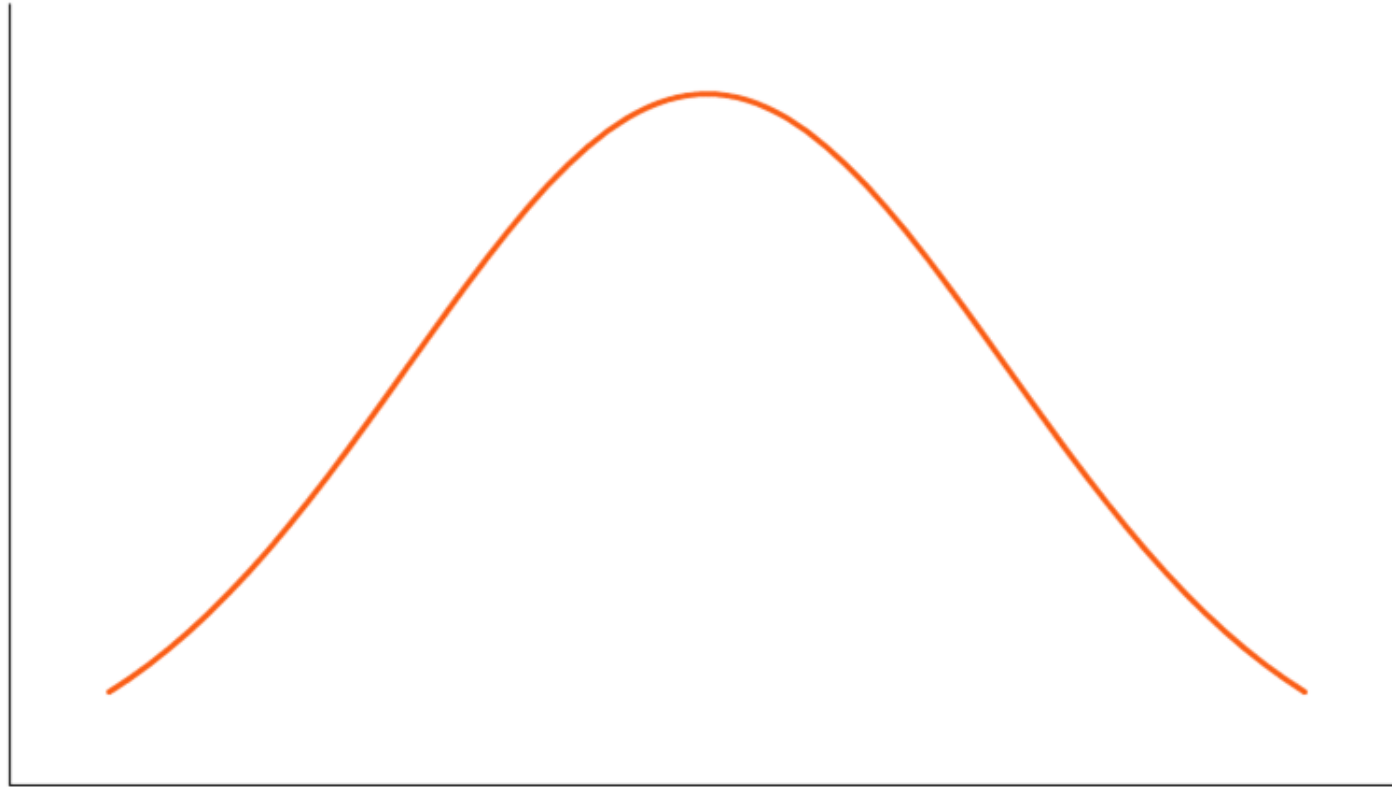
Leptokurtic

- Leptokurtic indicates a positive excess kurtosis. The leptokurtic distribution shows heavy tails on either side, indicating large [outliers](#). In finance, a leptokurtic distribution shows that the investment returns may be prone to extreme values on either side. Therefore, an investment whose returns follow a leptokurtic distribution is considered to be risky.



Platykurtic

- A platykurtic distribution shows a negative excess kurtosis. The kurtosis reveals a distribution with flat tails. The flat tails indicate the small outliers in a distribution. In the finance context, the platykurtic distribution of the [investment returns](#) is desirable for investors because there is a small probability that the investment would experience extreme returns.



Comparison of Different Kurtosis

